

Symmetry-Resolved Observable Contrast: Exact Invariant Witnesses and Data-Driven Order Parameters

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Abstract—The most discriminative observable may be physically inaccessible: it can depend on an arbitrary origin, violate a symmetry, or require an unrealizable instrument. We solve maximum observable contrast inside a finite-dimensional accessible observable algebra. If $\mathcal{E}_{\mathcal{A}}$ is the trace-preserving conditional expectation onto a unital $*$ -subalgebra $\mathcal{A}$, the largest expectation gap over effects in $\mathcal{A}$ is the trace of the positive part of $\mathcal{E}_{\mathcal{A}}(\rho_1 - \rho_0)$. Compact-group symmetry is the special case of twirling onto the commutant. Representation blocks then decompose distinguishability into physical sectors, while cyclic translation reduces exactly to Fourier power and admits $O(d \log d)$ updates. Controlled experiments verify the consequences. With only two training signals per class, translation-invariant contrast obtains 99.986% accuracy versus 50% for untwirled contrast and raw linear classification; a correctly specified Fourier-power classifier is perfect, isolating the value of the symmetry prior. A ten-qubit transverse-field Ising reduced-state response peaks at $g/J = 0.9625$, near the thermodynamic critical point 1, and resolves parity sectors. A Hückel difference density yields balanced attachment and detachment modes to 5.6×10^{-17} , and a simulated six-axis contact witness has 0.9993 overlap with the planted wrench screw. The framework unifies invariant testing, spectral diagnostics, optics, and density differences. Connections to gauge theory, quantum field theory, and holography are stated as concrete future calculations, not as a string-theory solution.

Index Terms—group invariance, interpretable physics, quantum state discrimination, symmetry-resolved analysis, trace distance

I. INTRODUCTION

AN unconstrained detector answers the wrong question when the observer has no absolute phase, the material is translation invariant, or only gauge-invariant readouts are physical. Data augmentation can approximate invariance, but it does not by itself identify the best realizable measurement or quantify which symmetry sector changed.

Invariant quantum hypothesis testing is established theory [1], and harmonic analysis provides the associated representation blocks [2], [3]. Symmetry-resolved entanglement likewise decomposes reduced states into charge sectors [4]. We combine these ideas with additive positive-state summaries to construct symmetry-/subalgebra-accessible observable contrast (SAOC). The central result is deliberately more general than a group: locality, block readout, bandwidth, and coarse graining can all be encoded as an accessible operator algebra.

This paper contributes:

- 1) a finite-dimensional conditional-expectation theorem for the optimal accessible bounded effect and its compact-group corollary;

Working paper. All numerical claims are backed by fixed-seed scripts, raw records, and manifests in the accompanying public repository. Chemistry and robotics results are controlled models, not deployment claims.

- 2) a sector decomposition that turns a scalar detector into a physical diagnostic, plus an exact fast Fourier case;
- 3) reproducible tests in signals, many-body physics, one-electron chemistry, and robot contact, including matched baselines and explicit negative boundaries;
- 4) a precise route toward gauge-invariant reduced-state comparison in quantum field and string-theoretic models without claiming such a calculation has already been completed.

II. ACCESSIBLE-OBSERVABLE THEORY

A. Unrestricted contrast

Let $\rho_0, \rho_1 \succeq 0$ have unit trace and $\Delta = \rho_1 - \rho_0$. Over all effects $0 \preceq E \preceq I$,

$$\max_E \text{Tr}(E\Delta) = \text{Tr}(\Delta_+) = \frac{1}{2} \|\Delta\|_1, \quad (1)$$

with $E^* = 1_{\Delta > 0}$. This is the operational trace-distance result of binary quantum discrimination [5]. We ask what remains when E must belong to the physical readout algebra.

B. Conditional expectation theorem

Let $\mathcal{A} \subseteq M_d(\mathbb{C})$ be a unital $*$ -subalgebra, and let $\mathcal{E}_{\mathcal{A}} : M_d \rightarrow \mathcal{A}$ be the trace-preserving conditional expectation (the Hilbert–Schmidt orthogonal projection onto $\mathcal{A}$).

Theorem 1 (accessible contrast).

$$\sup_{\substack{E \in \mathcal{A} \\ 0 \preceq E \preceq I}} \text{Tr}(E\Delta) = \text{Tr}[\mathcal{E}_{\mathcal{A}}(\Delta)_+]. \quad (2)$$

An optimizer is the positive support projector of $\mathcal{E}_{\mathcal{A}}(\Delta)$, formed inside $\mathcal{A}$.

Proof: For $E \in \mathcal{A}$, orthogonal projection gives $\text{Tr}(E\Delta) = \text{Tr}[E\mathcal{E}_{\mathcal{A}}(\Delta)]$. Conditional expectation preserves Hermiticity and $\mathcal{E}_{\mathcal{A}}(\Delta) \in \mathcal{A}$. Applying the Jordan bound within the finite-dimensional algebra proves the upper bound. Functional calculus in a $*$ -algebra places the positive support projector in $\mathcal{A}$, so the bound is attained. ■

The accessible distance is no larger than the unrestricted distance: coarse graining cannot increase distinguishability. Equality holds exactly when an unrestricted optimal effect has an equivalent representative in $\mathcal{A}$. This supplies a falsifiable measure of how much information an instrument constraint discards.

C. Group symmetry and sectors

For a compact group with unitary representation U_g , the commutant $\mathcal{A} = \{U_g\}'$ contains invariant observables. Its conditional expectation is the twirl

$$\mathcal{T}_G(X) = \int_G U_g X U_g^\dagger dg. \quad (3)$$

Thus,

$$\max_{\substack{0 \leq E \leq I \\ [E, U_g] = 0}} \text{Tr}(E\Delta) = \text{Tr}[\mathcal{T}_G(\Delta)_+]. \quad (4)$$

This one-shot finite-dimensional identity is consistent with the broader asymptotic study of symmetry-constrained quantum testing [1].

If

$$\mathcal{H} = \bigoplus_{\lambda} V_{\lambda} \otimes M_{\lambda}, \quad (5)$$

Schur's lemma gives

$$\mathcal{T}_G(\Delta) = \bigoplus_{\lambda} \frac{I_{V_{\lambda}}}{\dim V_{\lambda}} \otimes \Delta_{\lambda}. \quad (6)$$

Trace norm and positive gap add over λ . Each block is therefore a charge, parity, frequency, angular-momentum, or other representation-sector contribution, rather than an unnamed latent feature.

D. Cyclic translations

For the regular representation of C_d , the DFT diagonalizes every invariant operator. For a normalized signal x ,

$$\mathcal{T}_{C_d}(|x\rangle\langle x|) = F^\dagger \text{diag}(|Fx|^2)F. \quad (7)$$

The class state is the mean normalized power spectrum, and the optimal invariant effect selects positive spectral-contrast bins. Accumulation uses an FFT and a vector addition, $O(d \log d)$ time and $O(d)$ state, rather than a dense $d \times d$ matrix.

III. EXPERIMENTAL DESIGN

Every output has raw CSV, a JSON summary, a deterministic command, package versions, a Git identifier, and SHA-256 hashes. Unit tests compare FFT and explicit group twirls, invariant analytic and SDP optima, sector additivity, trace-distance contraction, Ising parity commutation, and Hückel density normalization.

A. Translation nuisance

The two classes are length-128 sinusoids at DFT bins 5 and 17. Every signal has random cyclic shift, random phase, Gaussian noise 0.8, and an exact negative partner. We vary training pairs per class from 1 to 16 and test on 600 signals per class over 12 seeds. Compared methods are raw linear logistic, untwirled rank-two contrast, invariant rank-two contrast, and a logistic classifier on the full Fourier-power representation.

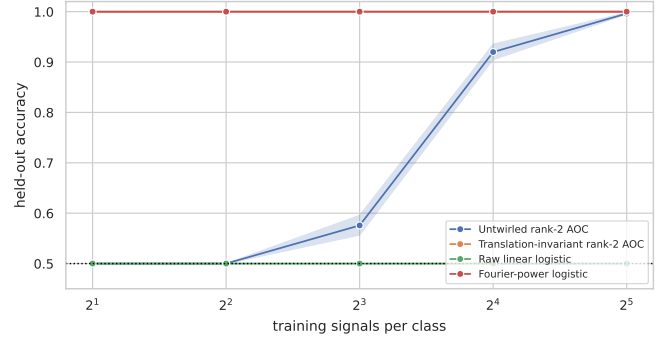


Fig. 1. Translation-nuisance sample efficiency. The invariant method and the correctly engineered Fourier baseline solve the task immediately; raw methods require enough shifts to estimate the orbit and linear means remain blind.

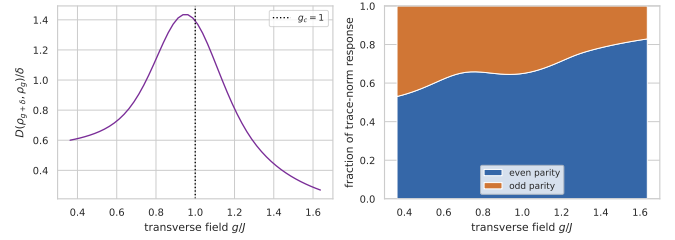


Fig. 2. Reduced-state trace-distance response and parity-sector decomposition in the ten-qubit transverse-field Ising model. The dotted line is the thermodynamic critical field.

B. Quantum phase response

We exactly diagonalize a periodic ten-qubit transverse-field Ising chain,

$$H = - \sum_i Z_i Z_{i+1} - g \sum_i X_i, \quad (8)$$

trace out five qubits, and compare adjacent reduced states on a field grid $\delta g = 0.025$. The subsystem parity $X^{\otimes 5}$ defines even and odd sectors. Trace-distance diagnostics are established tools for many-body states [6], [7]; the purpose here is sector-resolved witness validation, not discovery of an unknown model.

C. Difference-density chemistry

A half-filled ten-site Hückel ring is changed from uniform hopping to alternating hopping $1 \pm \delta$. The normalized occupied one-particle projectors are twirled over two-site translations. Positive and negative eigenvectors of their difference are natural attachment and detachment orbitals, an interpretation already used in molecular analysis [8], [9]. The example is a conservation and interpretability check, not predictive electronic structure.

D. Robot contact

A six-axis force/torque residual has zero mean. Contact adds a rank-one covariance along the physically valid wrench screw $s \propto [n^\top, (r \times n)^\top]^\top$. Exact sign pairing eliminates mean information. We learn a rank-one effect and compare it

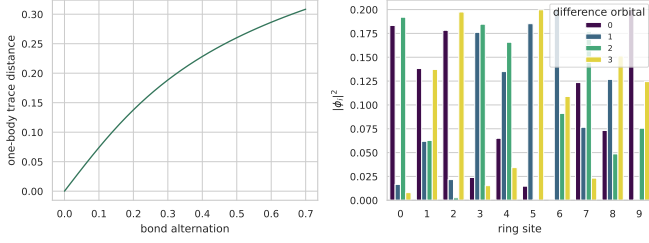


Fig. 3. Controlled Hückel result: total one-body state change and leading natural difference orbitals at maximum bond alternation.

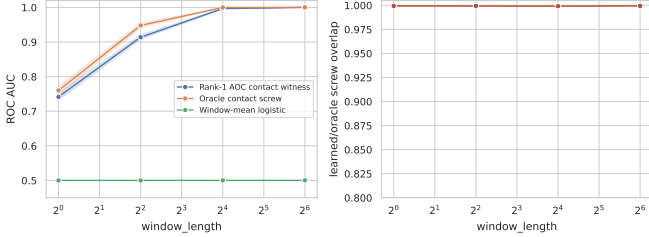


Fig. 4. Controlled robot-contact result. A low-rank quadratic witness recovers the planted wrench direction while the sign-paired mean is blind.

with the planted screw and window-mean logistic regression across window lengths 1, 4, 16, and 64, 12 seeds each. This is a structured simulation pending hardware or public dataset validation.

IV. RESULTS

At two training signals per class, Fig. 1 gives mean accuracy 0.999861 for invariant contrast, 1.0 for Fourier-power logistic, and exactly 0.5 for raw logistic and untwirled contrast. This is not evidence that SAOC beats an appropriate invariant feature map: they tie. It shows that quotienting an exactly known nuisance can transform the small-sample problem, while a learned orbit average needs more examples.

The trace-distance susceptibility in Fig. 2 peaks at $g/J = 0.9625$, close to the exact thermodynamic value 1. The maximum parity commutator residual is 3.54×10^{-9} . Finite-size displacement is expected; no critical exponent is inferred from one size. The result demonstrates that the learned bounded witness and its sector weights can serve as a data-driven response diagnostic alongside conventional order parameters [10].

At bond alternation 0.7, Fig. 3 gives one-body trace distance 0.30845. Positive attachment and negative detachment weights balance to 5.55×10^{-17} , as required by particle-number conservation. In real atom-centered nonorthogonal bases, the overlap metric and correlated density construction must be handled explicitly.

At 64-sample windows, both the learned and oracle contact scores in Fig. 4 have ROC AUC 1.0, whereas mean logistic has 0.5 by construction. Learned/oracle screw overlap averages 0.99925. The correct conclusion is recovery in a matched rank-one model, not robustness to payload, pose, friction, or sensor drift.

V. CONNECTIONS AND BOUNDARIES

A. Optics, vision, and atomistic representations

Normalized optical coherency matrices are density operators, so an accessible algebra can encode available wave plates, apertures, polarization analyzers, or spatial modes [11]. In vision and vibration, translation twirling gives an exact phase-blind power detector. It cannot distinguish homometric or phase-coded signals; bispectra, higher tensor states, or equivariant models are then necessary.

For molecular environments, rotations lead to spherical-harmonic blocks. SOAP [12] and the atomic cluster expansion [13] already provide systematic invariant representations. A credible use of SAOC is supervised, streaming selection of discriminative blocks from those established features, not replacement without molecular benchmarks.

B. Quantum field theory and string theory

The connection is a research program, not a solved application. Reduced states in quantum field theory can be compared by trace distance [6], and symmetry-resolved entanglement assigns contributions to conserved-charge blocks [4]. Given two computable reduced states $\rho_A(\theta)$ and $\rho_A(\theta + \delta)$, SAOC asks which bounded, gauge-invariant regional observable best distinguishes the couplings or backgrounds.

This question can be posed in lattice gauge theory, tensor networks, conformal field theory, and holography. The Ryu–Takayanagi relation connects boundary entanglement to bulk geometry [14], while the entanglement first law and modular Hamiltonian enter gravitational dynamics [15]. They are not identical to SAOC: a modular Hamiltonian $K = -\log \rho$ is a distinguished, generally unbounded state-dependent observable, whereas SAOC optimizes a bounded operational effect. Comparing their responses is meaningful; equating them without proof is not.

Gauge theories also need care because spatial Hilbert-space factorization and edge modes depend on the chosen observable algebra. A falsifiable first project is a small published lattice-gauge or matrix-product-state model: predeclare two couplings, form gauge-invariant reduced states, compute sector witnesses, and compare them with known flux or Wilson-loop diagnostics. Until that calculation exists, the repository makes no claim about a new string vacuum, duality, or quantum-gravity solution.

C. Failure modes

Twirling deliberately deletes asymmetry. If orientation or phase is the signal, invariant contrast is worse than the unrestricted method. Sector weights can be stable while eigenvectors inside a degenerate block are not identifiable. Dense non-Abelian decompositions can be expensive, and empirical state error can dominate small sector gaps. Physical instruments may impose a convex set that is not a $*$ -algebra; then Eq. (2) is replaced by a general convex optimization.

VI. CONCLUSION

Maximum observable contrast becomes physically meaningful only after the accessible measurements are specified. Conditional expectation projects the state difference onto that algebra; the positive part gives the exact optimal effect; and symmetry blocks reveal where distinguishability resides. The fast translation case, many-body parity response, difference-density orbitals, and contact wrench recovery show one operator principle across domains while preserving honest baselines. The strongest next tests are real streaming instruments and a concrete gauge-invariant many-body calculation.

REFERENCES

- [1] Fumio Hiai, Milán Mosonyi, and Masahito Hayashi. Quantum hypothesis testing with group symmetry. *Journal of Mathematical Physics*, 50:103304, 2009.
- [2] Jean-Pierre Serre. *Linear Representations of Finite Groups*. Springer, 1977.
- [3] Iman Marvian and Robert W. Spekkens. Modes of asymmetry: The application of harmonic analysis to symmetric quantum dynamics and quantum reference frames. *Physical Review A*, 90:062110, 2014.
- [4] Moshe Goldstein and Eran Sela. Symmetry-resolved entanglement in many-body systems. *Physical Review Letters*, 120:200602, 2018.
- [5] Carl W. Helstrom. *Quantum Detection and Estimation Theory*. Academic Press, 1976.
- [6] Jiaju Zhang, Paola Ruggiero, and Pasquale Calabrese. Subsystem trace distance in quantum field theory. *Physical Review Letters*, 122:141602, 2019.
- [7] Reyhaneh Khasseh, Jiaju Zhang, Markus Heyl, and M. A. Rajabpour. Identifying quantum many-body integrability and chaos using eigenstate trace distances. *Physical Review Letters*, 131:216701, 2023.
- [8] Daisuke Yamaki. Analysis method for comparison of two similar molecules using natural orbitals of the difference density matrix: Application to substituent effects. *Molecular Physics*, 113:342–347, 2015.
- [9] Andrew J. Bovill, Ali Abou Taka, Hassan Harb, and Hrant P. Hratchian. Excitation/relaxation analysis of electronic transitions using difference density natural orbitals. *Journal of Chemical Theory and Computation*, 22(2):930–939, 2026.
- [10] Juan Carrasquilla and Roger G. Melko. Machine learning phases of matter. *Nature Physics*, 13:431–434, 2017.
- [11] Leonard Mandel and Emil Wolf. *Optical Coherence and Quantum Optics*. Cambridge University Press, 1995.
- [12] Albert P. Bartók, Risi Kondor, and Gábor Csányi. On representing chemical environments. *Physical Review B*, 87:184115, 2013.
- [13] Ralf Drautz. Atomic cluster expansion for accurate and transferable interatomic potentials. *Physical Review B*, 99:014104, 2019.
- [14] Shinsei Ryu and Tadashi Takayanagi. Holographic derivation of entanglement entropy from the anti-de sitter space/conformal field theory correspondence. *Physical Review Letters*, 96:181602, 2006.
- [15] Nima Lashkari, Michael B. McDermott, and Mark Van Raamsdonk. Gravitational dynamics from entanglement thermodynamics. *Journal of High Energy Physics*, 2014(4):195, 2014.